Problem 1: Find the Promos. A screenplay-to-promo finder

[Name] · [Roll no.] · [Branch] · Prototype: [repo link] · Raw outputs: [Drive link] · Video: [optional]

Why this problem

It is the one with a real input pack and a checkable output: every promo is a set of timestamps an editor can open and judge in seconds. The pain is volume. An editor watches every episode of every weekly show, but the screenplays already hold most of what that editor looks for (who is cruel to whom, which line is shouted, where the writer put the cliffhanger sting). So the tool reads the script, not the video, and hands the editor a ranked shortlist to check rather than 50 episodes to watch.

Assumptions (the brief was silent)

- Timestamps are **episode-relative** and the script timings match the final cut closely enough to pick cut points; the editor trims frames. Script ≠ footage: a beat that reads big may play small.
- **Hook = a beat that starts in the first 3 s of the clip**; cliffhanger = the last ~5 s is unresolved. Clips are 20–60 s in total; 25–50 s is preferred. A promo may join up to 3 segments, always in story order.
- An ad viewer who taps lands on **Ep 1**. Promos from early episodes therefore convert and spoil less, so they post first. Late-twist promos go to retargeting. The show title already gives away "He was a Duke".
- Same script format for every show (CAMERA / ACTION / SOUND FX / CHARACTER (tags) + Start/End). New formats need only a new parser.

How it works: code does the searching, AI does the taste, a human signs off

Step	Who does it	What happens
1 Parse	Code	3,727 timestamped events from 50 episodes; CAMERA blocks give the shot cuts.
2 Generate	Code	Every 20–60 s window plus jump cuts within an episode, Ep N cliffhanger + Ep N+1 payoff, and a 3-beat *premise montage* built from the title (setup → "married" → "Duke"). Cuts snap to shot boundaries and never split a line.
3 Score	Rules (free)	Hook 35%, cliffhanger 30%, standalone 15%, peak intensity 10%, pace 10%. Signals: writer's delivery tags (furious, gloating…), violence/danger/betrayal/reveal words, open questions, threats, arrivals, a blow about to land, an unanswered cruel line, the writer's cliffhanger cue; penalties for resolution words, context openers, >4 speakers, dead air.
4 Re-rank	LLM (optional)	Reviews the top 40 and may tighten the cut, but can only answer with **event IDs it was shown**. IDs are validated, re-snapped and kept within 20–60 s, else it falls back to the rule result. It cannot invent timestamps or quotes. Model is one flag: `--llm gemini | groq | openai | anthropic | ollama | openai_compat`; responses are cached.
5 Rank + QC	Code	Score − QC penalties + funnel bonus; diversity ordering; near-duplicate cuts become "alternate cuts". Flags per promo: weak hook/cliffhanger, may resolve, needs context, dead air, spoiler risk.
6 Approve	Editor	Picks from the shortlist in a Streamlit UI or CSV, watches the footage, trims frames.

What it found (rules only, full pack, 3.7 s, 260 candidates → 25-promo shortlist)

#	Segments	Hook (first 3 s)	Cliffhanger (cut on)
9	Ep3 00:58.20–01:28.10	"Guards! Remove her!" She is dragged from the church	Stranger drops a guard → "He's only one man!"
4	Ep12 00:26.80–01:09.50	"You planted it!"	"Fail, or refuse, and your husband hangs from the gallows."
1	Ep24 00:32.80–01:10.50	Riding-crop strike	"Do you want the same punishment?"
3	Ep32 00:52.50–01:22.50 + 02:14.80–02:32.00	"You are lying!"	Red-hot iron stops millimetres from her eye
10	Ep36 02:25.80–03:17.10	"What wedding?"	"…their dead Duke is coming home."
13	Ep26 01:50.80–02:00.88 + Ep27 00:44.20–01:20.80	"Where did you get this ring?"	Doors burst open; she is framed
20	Ep1 00:39.00–00:50.40 + Ep12 01:05.20–01:14.00 + Ep45 00:16.90–00:24.50	"Because he is marrying me."	"You're the Duke?" (premise montage)

Post first: Ep3, then Ep1 and Ep12. They are early (the viewer lands where the promo happens), play cold (sister, husband and gallows are named on screen) and end mid-threat. Then Ep24, the Ep26→27 ring reveal and the premise montage. Hold Ep45 and the Ep44→45 reveal for viewers already watching. The tool's own order puts Ep24 first (#1), which is fine. In my review below, Ep3 (#9) is the better opener.

How I know it's good, and where it picks badly

- **Recall vs a reference list.** Before seeing any output I hand-picked 17 promo moments from the scripts. The tool ranks **7/17 in its top 10, 10/17 in the top 20, 12/17 in the top 40**, and all 17 appear somewhere in the 260 candidates.
- **Cliffhanger detector vs the writers.** 38 episode endings have a clear writer cue ("cliffhanger sting" vs "warm fade-out"). Using dialogue only (SFX hidden, as on a script without music notes), the detector gets **precision 0.94, recall 0.53**. When it says cliffhanger it is almost always right, but it misses visual ones (Ep1 ends on her face hardening).
- **Manual review of the top 20** (from the script text, not footage): **10 post-ready, 7 right moment with a wrong cut, 3 reject**. Failure patterns:

Where it picks badly	Example	Why / fix
Loud beats beat meaningful ones	Ep1 starts 7 s after the real hook "Because he is marrying me."	Tags and "!" drive the hook score → LLM judge

Where it picks badly	Example	Why / fix
Exposition reads as stakes	Ep43: "Arrest him!" then a Crown official reads records (rejected)	No exposition penalty → LLM "needs backstory" check
60 s cap cuts before the payoff	Ep11 cuts on "Happy?", 25 s before the planted watch is found	Should jump-cut to the payoff
Visual spectacle under-ranked	Fire (Ep18), ledger thrown in mud (Ep5), explosion (Ep41), proposal (Ep3): ranks 51–79	Text can't see scale → LLM / video signals
Same scene twice	3 Ep32 promos in top 20; 2 share an ending	Raise episode-diversity penalty
Word-list labels	"You have thrown yourself away!" labelled a threat	Labels are hints; the LLM rewrites them

Honest limits. The rules were tuned on this one show, so these numbers are optimistic; the real test is the new show in the interview. The reference list and the review are one person's judgement, not editors' or ad performance. The LLM layer is built and tested (stub HTTP server; invented IDs are rejected) but **not yet measured**: the build machine could not reach any model API. I will run it with a free Gemini key and compare its top 20 against the same review.

Scale and cost

- Rules: ~4 s and $0 per 50-episode show on a laptop, deterministic, no limits. Parsing is the only show-specific part.
- LLM pass: 40 reviews × ~1.1k tokens ≈ 46k input + 7k output per show → about **$0.03/show** on Gemini 2.5 Flash ($0.30/$2.50 per M), $0.007 on Flash-Lite, ~$0.18 on a pro-class model. At 5 shows a week that is under $50 a year even on the top tier. The free tier's rate limit is handled by backoff and an on-disk cache.
- Real-world loop: log which shortlisted promos editors actually post and their CTR / cost per install, then re-weight the five scores (or few-shot the LLM with winners). That replaces my judgement with Kuku's data. The same pipeline works per language when scripts are localised.

AI tools used, and where they got it wrong

Claude (via Claude Code) helped me read all 50 scripts, write the parser/scorer/UI and draft this doc. The decisions above are mine: weights, funnel-first ranking, ID-only LLM answers, the evaluation design. Where it got it wrong: (1) its first LLM integration blended LLM-reviewed and unreviewed promos on different score scales, so the reviewed ones sank. A mock-model run caught it, and reviewed promos are now ranked as their own tier. (2) Its violence word list tagged figurative speech ("thrown yourself away") as physical violence. Violent words in dialogue now count as threats, but the labels remain approximate.